MILITARY SPECIFICATION

FUZE, BASE DETONATING, M578 LOADING, ASSEMBLING AND PACKING FOR RENOVATION

This Specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Specification covers the renovation and packing for one type of artillery fuze designated as Fuze, Base Detonating, M578.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of this Specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-116 - Preservation, Methods of

STANDARDS

MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-453 - Radiographic Inspection

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARMY

9323961 - For Renovation Only
Fuze, Base Detonating
M578, Loading Assembly
9323905 - Radiographic Procedures for Inspection,
Projectile, 105MM, HEP-T, M393A2
9323955 - Cartridge, 105MM, HEP-T, M393A2, Renovated
Assembly and Marking Diagram
9323980 - Envelope Dwg. for Spin Tester, Arming
Delay Assembly, Fuze M578, Renovation
8860738 - Locking Segment and Striker Needle Assembly
8860752 - Arming Delay Assembly, Loaded
8796522 - Marking Diagram and Sealing
for Wooden Packing Boxes
8887101 - Box, Fiber for Fuze, BD, M578
8887102 - Box, Packing, Ammunition for Fuze,
BD, M578

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Fuze Assembly.-The fuze assembly shall comply with all requirements specified on Drawing (dwg) 9323961 and with all requirements specified in applicable specifications.

3.3 Waterproofness.-The fuze assembly shall show no evidence of moisture on the interior surface after the test specified in 4.5.1.

3.4 Radiographic Examination (See 4.4.2.9).-The fuze assembly shall be radiographically examined as specified in 4.5.4, and shall show no evidence of the following:

a. Armed fuze (shutter in armed position)
b. Missing component
c. Mis-assembled component
d. Damaged component
e. Missing, cracked or incomplete explosive charge

3.5 Workmanship.-All parts shall be dry and free of burrs, cracks, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used. All required marking shall be neat and sharply defined.

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3.6 Proving Ground

3.6.1 Arming Safety at 70° ± 2.5°F (21.1° ± 1.4°C).-The fuze (dwg. 9323961) shall not function when impacted against a four (4) inch (10.2cm) thick plywood screen placed 16 feet (4.88 meters) from the muzzle of the weapon.

3.6.2 Functioning at 70° ± 2.5°F (21.1° ± 1.4°C).-The fuze (dwg. 9323961) shall function when impacted against a four (4) inch (10.2cm) thick plywood screen placed 150 feet (45.7 meters) from the muzzle of the weapon.

3.7 Spin Tests

3.7.1 Arming Delay Assembly-Arming, Non-Arm and Arming Time.-When the loaded arming delay assembly is spun about its axial center to the specified speeds, the assembly shall be armed or disarmed with arming times as follows:

RPM	SHUTTER POSITION	ARMING TIME
6,500 Min	Disarmed	
9,000 - 100	Armed	.036 to .080 Sec.
15,000 - 200	Armed	.020 to .040 Sec.

NOTE: Alternate Timing Requirement. The Arming Time Test at 9,000 RPM is an alternate to the test at 15,000 RPM.

3.8 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are part of the detailed specification.

4.2 Classification of Inspections.-The following types of inspections shall be conducted on this item:

a. First Article on renovated fuzes
b. Production Quality Conformance Inspection on renovated fuzes

4.3 First Article Inspection.-These tests are to be conducted to assure the quality of the renovated fuze prior to assembly into the reworked configuration of Cartridge, 105MM, HEP-T, M393A2 Lots.

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4.3.1 Submission.-The renovator of the fuzes, prior to the initiation of regular production, shall ship 150 each, of the following components: Ring, Locking (dwg. 9323864), Cup, Lead, Booster (dwg. 9323863), and Cap (dwg. 9323865) to a Government approved facility as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. An additional 150 renovated fuze assemblies shall be shipped to Milan Army Ammunition Plant for assembly into the complete rounds (see 4.3). The assembled rounds shall then be shipped to the appropriate Proving Ground for acceptance testing, Arming Safety (see 3.6.1 and 4.5.5.1) and Functioning (see 3.6.2 and 4.5.5.2).

4.3.2 Inspections to be Performed.-The First Article sample assemblies and components shall be inspected for conformance with any or all of the requirements of this specification including proving ground tests, and any or all of the requirements of the applicable drawings (see Table I).

4.3.3 Rejection.-See MIL-A-48078 (PA). In the ballistic tests, if a premature burst occurs or if one or more fuzes fail to comply with the requirements of 3.6.1, the initial production sample shall be rejected. In addition, if two or more fuzes fail to comply with 3.6.2, the initial production lot shall be rejected.

4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Comp A-5 for lead charges from not more than two consecutive lots with the same lot interfix number from one manufacturer.

c. Comp A-5 for the booster pellets from not more than two consecutive lots with the same lot interfix number from one manufacturer.

d. The integrity of each fuze lot prior to and following renovation shall be maintained.

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4.4.2 Examination.-One hundred percent examination shall be performed for all critical defects and major defects when so specified. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plans of MIL-STD-105, Level II. All non-conforming material shall be rejected, see MIL-A-48078.

NOTE: Critical Defect Reporting.-A report of all Critical Defects found, number of parts or assemblies inspected for critical defects, and corrective action taken, if any shall be compiled monthly for the duration of the contract and copies forwarded to the Contracting Officer.

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TABLE I - FIRST ARTICLE SAMPLE

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Renovated Fuze, Base Detonating, Parts and Loaded Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze, Loading Assembly (Dwg. 9323961)				
	Radiographic examination	150	(a)	3.4	4.5.4
	Arming Safety Test	75	(b)	3.6.1	4.5.5.1
	Functioning Test	75	(b)	3.6.2	4.5.5.2
	Arming Delay Assembly, Loaded (Dwg. 8860752)				
	Non-Arming Test	150	(a)	3.7	4.5.7
	Arming Test	150	(a)	3.7	4.5.7
	Arming, Delay Time Test	150	(a)	3.7	4.5.7
	Cup, Booster, Lead (Dwg. 9323863)				
	Examination for defects	150	(a)	3.2	4.4.2.2
	Ring, Locking (Dwg. 9323864)				
	Examination for defects	150	(a)	3.2	4.4.2.3
	Cap (Dwg. 9323865)				
	Examination for defects	150	(a)	3.2	4.4.2.4

NOTES: a. To be performed at contractor's plant
b. At Proving Ground

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Assembly, Arming Delay		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1.	Acceptance mark on ass'y missing			3.2	Visual
Major	None defined				
Minor 201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Booster Pellet		9276679
			NEXT HIGHER ASSEMBLY
			9323961

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1.	Pellet cracked			3.2	Visual
Major 101	Density measurement test	10		3.2	4.5.2
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Booster, Lead Cup				9323863
					NEXT HIGHER ASSEMBLY
					9323961
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of bottom		0.40%	3.2	Gage
102	Flatness of base		0.40%	3.2	Gage
Minor					
201	Pitch diameter of thread		0.65%	3.2	Gage
202	Major diameter of thread		0.65%	3.2	Gage
203	Diameter of lead charge cavity		0.65%	3.2	Gage
204	True position of lead charge cavity with thread		0.65%	3.2	Gage
205	Depth of pellet cavity		0.65%	3.2	Gage
206	Diameter of pellet cavity		0.65%	3.2	Gage
207	True position of pellet cavity with thread		0.65%	3.2	Gage
208	True position of thread pilot at base end		0.65%	3.2	Gage
209	True position of thread pilot at open end		0.65%	3.2	Gage
210	Overall length		0.65%	3.2	Gage
211	Diameter of thread pilot at open end		0.65%	3.2	Gage
212	Length of thread pilot at base end		0.65%	3.2	Gage
213	Length of thread pilot at open end		0.65%	3.2	Gage
214	Width across wrench flats including true position		0.65%	3.2	Gage
215	Chamfer missing or incorrect		0.65%	3.2	Visual
216	Surface finish		0.65%	3.2	Visual/Gage
217	Radii missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	CONTINUATION SHEET				9323863
					NEXT HIGHER ASSEMBLY
					9323961
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
218	Protective finish missing or damaged		0.65%	3.2	Visual
219	Evidence of poor workmanship		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63253 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Ring, Locking		9323864
			NEXT HIGHER ASSEMBLY 9323961

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pitch diameter of thread		0.65%	3.2	Gage
202	Minor diameter of thread		0.65%	3.2	Gage
203	Outside diameter		0.65%	3.2	Gage
204	True position of outside diameter with pitch diameter of thread		0.65%	3.2	Gage
205	Length		0.65%	3.2	Gage
206	Length to base of wrench slot		0.65%	3.2	Gage
207	Length of counterbore		0.65%	3.2	Gage
208	Diameter of counterbore		0.65%	3.2	Gage
209	Width of wrench slots		0.65%	3.2	Gage
210	Chamfer missing or incorrect		0.65%	3.2	Visual
211	Protective finish missing or incomplete		0.65%	3.2	Visual
212	Surface finish		0.65%	3.2	Visual
213	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Cap				9323865
					NEXT HIGHER ASSEMBLY 9323961
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of top		0.40%	3.2	Gage
Minor					
201	Pitch diameter of thread, min.		0.65%	3.2	Gage
202	Major diameter of thread		0.65%	3.2	Gage
203	True position of outside diameter with pitch diameter of thread		0.65%	3.2	Gage
204	Outside diameter		0.65%	3.2	Gage
205	Length		0.65%	3.2	Gage
206	Diameter of thread relief		0.65%	3.2	Gage
207	Radii missing		0.65%	3.2	Visual
208	Surface finish improper		0.65%	3.2	Visual/Gage
209	Protective coating missing or damaged		0.65%	3.2	Visual
210	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Booster Lead Cup, Prior to Inserting Booster Pellet				9323961
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1.	Lead charge above flush			3.2	Gage
Major 101	Lead charge missing, or more than max below flush		0.40%	3.2	Visual/Gage
Minor	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Booster, Lead Cup - Prior to Assembling to Fuze				9323961
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Waxed felt closing disc missing			3.2	Visual
Major					
101	Flatness of bottom surface of booster lead cup		100%	3.2	Gage
102	Booster pellet missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9323961
4.4.2.8	Fuze Assembly, Prior to Assembling Booster Lead Cup Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Shutter in armed position			3.2	Visual
2.	Striker length excessive			3.2	Gage
3.	Spin test acceptance marking missing or reject mark present on Arming Delay Assembly			3.2	Visual
Major					
101	Any component missing or misassembled		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9323961
4.4.2.9	Fuze, Assembly				NEXT HIGHER ASSEMBLY 9323955
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Shutter not in safe position	(a)		3.4	Test 4.5.4
2.	Delay assembly missing or misassembled	(a)		3.4	Test 4.5.4
3.	Creep spring missing	(a)		3.4	Test 4.5.4
4.	Radiographic inspection acceptance mark missing	(a)		3.4	Visual
Major					
101	Segment missing or improperly assembled	(a)	100%	3.4	Test 4.5.4
102	Booster pellet cracked	(a)	100%	3.4	Test 4.5.4
103	Component damaged, missing or misassembled	(a)	100%	3.4	Test 4.5.4
104	Lead booster cup fails min assembly torque	(b)	0.40%	3.2	Gage
105	Cap fails min assembly torque	(b)	0.40%	3.2	Visual
106	Alignment screw above flush		0.40%	3.2	Visual
107	Length, max.		0.40%	3.2	Gage
108	Thread damaged		0.40%	3.2	Visual
109	Adhesive missing		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES: (a) Radiographic examination 4.5.4
(b) Torque Test - See 9323961

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Box, Fiber Packing, Prior to Sealing		8887101
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any packing material missing		0.65%	3.5	Visual
202	Separator damage to extent that fuzes touch each other		0.65%	3.5	Visual
203	Fuze missing		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887101
4.4.2.11	Sealed, Box, Fiber				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal box completely		0.65%	3.5	Visual
202	Carton torn, cut or punctured		0.65%	3.5	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Contents loose		0.65%	3.2	Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887101
4.4.2.12	Sealed Barrier Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OF .4%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Seal improper or incomplete		0.40%	3.5	Visual
102	Bag punctured, torn or cut		0.40%	3.5	Visual
103	Heat seal test			3.2	4.5.3
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887102
4.4.2.13	Packing Box, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Carton missing		0.40%	3.5	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63253 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887102
4.4.2.14	Packing Box, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL (%)	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken, board loose or protruding nail		0.40%	3.5	Visual
102	Strapping missing, broken or loose		0.40%	3.5	Visual/Manual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.5	Visual
202	Strapping improperly engaged or mislocated		0.65%	3.5	Visual
203	Contents loose		0.65%	3.5	Visual/Manual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing

4.4.3.1 Waterproofness (see 3.3) - Major Defect

4.4.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, 50 fuze assemblies (see dwg. 9323961) shall be selected from each lot for this test. If two or more assemblies fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.1.

4.4.3.1.2 After three consecutive lots have complied with the criteria of 4.4.3.1.1, 15 sample fuze assemblies shall be selected from each lot for this test. If one or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.1.

4.4.3.2 Proving Ground (see 3.6) - Fuze Assembly 9323961 Assembled Into Complete Round M393

4.4.3.2.1 Arming Safety (3.6.1) - Critical Defect.-A sample of ten (10) fuzes shall be randomly selected from each lot for this test. If one or more assemblies fail to comply with the specified requirement the lot shall be rejected. The test shall be performed as specified in 4.5.5.1.

4.4.3.2.2 Functioning (3.6.2) - Major Defect.-A sample of ten (10) fuze assemblies shall be selected from each lot for this test. If two or more assemblies fail to comply with the specified requirement the lot shall be rejected. The test shall be performed as specified in 4.5.5.2.

4.4.3.2.2.1 If one fuze fails to function as specified in 3.6.2, then an additional sample of 65 fuzes shall be loaded into complete rennovated rounds and tested for functioning. If one fuze, from this additional sample, fails to function, then the lot shall be rejected.

4.4.3.2.3 First Article Test.-Seventy five fuzes shall be tested for arming safety and functioning as specified in 4.5.5.1 and 4.5.5.2. If one or more fuzes function during the arming safety phase; or, if two or more fuzes fail to function upon impact during the functioning phase, the first article sample shall be rejected.

4.4.3.3 Arming Delay Assembly (see dwg. 8860752)

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4.4.3.3.1 Arming and Arming Delay Time (see 3.7.1)

Major Defect - Arming.
Major Defect - If arming time is greater than the maximum specified.
Critical Defect - If arming time is less than the minimum specified.

These tests shall be performed 100 percent. The tests shall be performed as specified in 4.5.7, using test equipment specified in 4.4.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein, is described in the Paragraph Reference/Inspection method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.4 herein.

4.5 Test Methods and Procedures

4.5.1 Waterproofness.-The fuze assemblies shall be immersed in water to a minimum depth of six (6) inches for a period of not less than four (4) hours. At the end of immersion time, the assemblies shall be removed from the water, wiped dry and disassembled to determine compliance with the requirement.

4.5.2 Density of Booster Pellet.-The density of the booster pellet shall be determined in accordance with Method 203.1 of Standard MIL-STD-650. If any pellet fails to comply with the requirements, the lot shall be rejected.

4.5.3 Heat Seal of Barrier Bag (see dwg. 8887101).-Test procedures and sampling acceptance criteria for this test shall be as specified in Specification MIL-P-116 for the "Heat Seal Test".

4.5.4 Radiographic Inspection (see 6.2).-Prior to assembly into projectiles, all fuzes shall be positioned for radiographic exposure, such that complete coverage of one or more fuzes is provided on one film. The alignment of fuzes shall be controlled such that the radiation source shall lie in a plane perpendicular to the longitudinal axis of each fuze and intersecting the fuze at its center. Orientation of each fuze shall

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register on the external set screw, such that the set screw shall project onto the central axis of the radiographic fuze image (zero degree position). Source to film distance shall be not less than four (4) feet and peak radiation energy shall be not less than 300 kilovolts. Radiographic technique shall be maintained to consistently produce image quality characteristics displayed on approved standard reference radiographs (see 6.2), in accordance with radiographic sensitivity procedures specified herein. Radiographs of all fuzes shall be examined by qualified personnel for all radiographic inspection requirements of MIL-C-63249, Projectile, 105MM, HEP-T, M393A2. Any fuze showing one or more specified defects (see 3.4) shall be classed defective and removed from the lot. Any fuze determined to be defective by radiographic inspection, shall be re-radiographed for verification of the defect or defects. Fuzes with verified critical defects shall be placed in a holding area (bond) and the contractor shall immediately report the discovery to the Contracting Officer.

4.5.5 Proving Ground (see 3.6).-Fuze Asssmbly Dwg. 9323961 Assembled to Complete Round M393 in accordance with dwg. 9323955.

4.5.5.1 Arming Safety.-The loaded rounds (M393) shall be fired through a four (4) inch (10.2cm) thick wood bursting screen placed the required distance from the gun. The rounds shall be observed for compliance with the applicable requirement.

4.5.5.2 Functioning.-The loaded rounds shall be fired as specified in 4.5.5.1 except that the screen shall be placed at the functioning distance from the gun. The rounds shall be observed for compliance with the applicable requirements.

4.5.6 Loaded Arming Delay Assembly (see 3.7.1).-Since the arming delay assembly contains a loaded detonator, adequate safety precautions shall be taken to protect personnel handling and testing the assembly.

4.5.6.1 Non-arming.-The arming delay assembly shall be placed in a suitable holding device that will apply the compressive load. The holding device, with the arming delay assembly, shall then be spun at the required non-arming speed and observed for compliance with the applicable requirement (see 3.7.1).

4.5.6.2 Arming.-The loaded arming delay assembly shall be tested as specified in 4.5.6.1 except that the rotation shall be at the required arming speed.

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4.5.6.3 Arming Delay Time.-When the arming time interval is determined by the low or high speed rotation method, the loaded arming delay assembly shall be tested in a device similar to that specified on dwg. 9323980, to time the interval between start of rotation and the engagement of the locking plate in the recess. For both the low speed and the high speed rotation, the rotating device must be modified with a mechanism to restrain the safety plunger until the specified rotation speed is attained. The timer shall time the interval between release of the loaded shutter assembly by the safety plunger and the engagement of the locking plate in the recess. The loaded arming delay assembly shall be observed for compliance with the requirement. As a minimum, the spin equipment shall be calibrated at the beginning of each eight (8) hour shift and at the end of each eight (8) hour shift by testing of Standards (see dwg. 9323980) in the spin equipment. If the standard test is found unacceptable all accepted Delay assemblies tested since the last acceptable calibration shall be re-inspected 100 percent.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C - The fuze assemblies shall be preserved and packaged in accordance with dwg. 8887101.

5.2 Packing

5.2.1 Level C - The packaged fuze assemblies shall be packed in accordance with dwg. 8887102.

5.3 Marking.-Marking shall be as specified on dwgs. 8887101 and 8887102.

6. NOTES

6.1 Intended Use.-The fuze assembly covered by this specification is intended for use on the renovated M393A2 complete round "only".

6.2 Radiographic Examination (see 4.5.4).-Radiographic inspection shall be performed only by personnel certified in accordance with practices and criteria in accordance with the following:

a. US Army DARCOM personnel shall be certified in accordance with DARCOM Regulation 702-22.

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b. Other Government and Contractor personnel shall be certified in accordance with Standard MIL-STD-410, Level II.

6.2.1 Radiographic Film.-Only an Industrial Radiographic Film of an approved type shall be used to radiograph the fuze. Substitutes may be used only after evaluation and approval by ARRADCOM, Dover, N.J. 07801. The following are initially approved types:

a. Eastman Kodak Company, Types AA, T, and M.
b. Dupont DeNemours & Co., Types NDT-55, NDT-65, and NDT-70.
c. GAF - 800.
d. GAF - 400.

6.2.2 Item Identification.-An identification numbering system shall be employed which shall provide direct, unambiquous correlation between each item and its radiographic image, and which shall be used for reporting purposes.

6.2.3 Radiographic Sensitivity.-The sensitivity of the radiographic technique for fuze inspection shall be determined by examination of internal fuze features. Two (2) or more Standard Reference Radiographs of fuzes, in required orientation, shall be produced and used for comparison. Reference radiographs shall be submitted to Commander, ARRADCOM, Dover, N.J. 07801, ATTN: DRDAR,QAR-E for review and approval. The Radiographic Sensitivity shall not be considered satisfactory unless all major internal features of every fuze image are as sharply defined as on the approved reference radiographs. Failure to meet this sensitivity requirement on any fuze shall require re-radiographing of that fuze and a diagnosis of that failure.

6.3 Ordering Data.-In addition to the requirements of MIL-A-48078, procurement documents shall specify the following as a minimum.

a. Provisions for Government verification inspection.
b. Contractor warranty of fuze for 180 days (not required for GOCO plant contract).
c. Critical Defect reporting as required by 4.4.2.
d. Resubmission of first article samples when a lapse in production or a significant change in the manufacturing process occurs.

6.4 Submission of Designs for Approval (see 6.2 of MIL-A-48078).-Submit equipment design, as required to Commander, ARRADCOM, Hqt, Dover, N.J. 07801, ATTN: DRDAR-QAR-E.

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6.5 Submission of Test Data and Ammunition Data Cards.- In addition to the normal distribution of records when the renovated fuze is procured by the Army Armament Materiel Readiness Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARRADCOM, ATTN: DRDAR-LCR-E and DRDAR-QAR-E, Dover, N.J. 07801; and, Commander, ARRCOM, ATTN: DRSAR-QAD, Rock Island, Ill. 61201.

Custodian:
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Preparing Activity:
Army - AR

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-F-63253 Fuze, Base Detonating, M578 LAP For R4novation

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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